Exercitii

ex1 Să se determine transformatele Laplace pentru următoarele funcții originale

a) $f(t)=\int_0^t \frac{\sin \tau}{\tau} d\tau$

$$f(t)=\int_0^t \frac{\sin \tau}{\tau} d\tau \xrightarrow[\text{orig.}]{\text{integ.}} \frac{\mathcal{L}\left[\frac{\sin t}{t}\right](p)}{p} \xrightarrow[\text{imag.}]{\text{integ.}}$$

$$=\frac{1}{p}\int_p^\infty \mathcal{L}[\sin t](v)\,dv=\frac{1}{p}\int_p^\infty \frac{1}{v^2+1}dv=\frac{1}{p}\operatorname{arctg} v\Big|_p^\infty$$

$$=\frac{1}{p}\left(\frac{\pi}{2}-\operatorname{arctg} p\right)$$

b) $f(t)=(t-1)^2\sin(t-1)$

$$f(t) \xrightarrow[\text{întârz.}]{\mathcal{L}} e^{-p}\mathcal{L}[t^2\sin t](p) \xrightarrow[\text{imag}]{\text{deriv}} e^{-p}\mathcal{L}^{II}[\sin t](p)$$

$$=e^{-p}\left(\frac{1}{p^2+1}\right)''=e^{-p}\left(\frac{0-2p}{(p^2+1)^2}\right)'=e^{-p}\frac{-2(p^2+1)^2-(-2p)\cdot 2(p^2+1)\cdot 2p}{(p^2+1)^{4}}$$

$$=e^{-p}\frac{6p^2-2}{(p^2+1)^3}$$

c) $f(t)=\frac{1-\cos t}{t}$

$$f(t) \xrightarrow[\text{imag.}]{\text{integ.}} \int_p^\infty \mathcal{L}[1-\cos t](v)\,dv \xrightarrow{\text{liu.}} \int_p^\infty \frac{1}{v}-\frac{v}{v^2+1}dv=$$

$$= \left[\ln v - \frac{1}{2} \ln(v^2+1) \right] \Big|_p^\infty = \ln \frac{v}{\sqrt{v^2+1}} \Big|_p^\infty = -\ln \frac{p}{\sqrt{p^2+1}}$$

d) $f(t) = te^{3t} \operatorname{sh}(5t)$

$$f(t) \xrightarrow[\text{deplas}]{\mathcal{L}} \mathcal{L}[t \cdot \operatorname{sh}(5t)](p-3)$$

$$\mathcal{L}[t \operatorname{sh}(5t)](p) \xrightarrow[\text{imag}]{\text{der u}} -\mathcal{L}'[\operatorname{sh}(5t)](p) = -\left(\frac{5}{p^2-25}\right)' = -5 \cdot \frac{-2p}{(p^2-25)^2} = \frac{10p}{(p^2-25)^2}$$

$$\Rightarrow \mathcal{L}[t \cdot \operatorname{sh}(5t)](p-3) = \frac{10(p-3)}{[(p-3)^2-25]^2}$$

ex2 Să se determine transformata Laplace pentru următoarea funcție

$$f(t) = \begin{cases} 2, & 0 < t < 1 \\ \frac{t^2}{2}, & 1 < t < \pi/2 \\ \cos t, & t > \pi/2 \end{cases}$$

$$f(t) = 2(\underset{\downarrow}{1} - u(t-1)) + \frac{t^2}{2}\left(u(t-1) - u\left(t - \frac{\pi}{2}\right)\right) + \cos t \cdot u(t-\pi/2)$$

$u(t)$

$$f(t) = 2 - 2u(t-1) + \frac{(t-1+1)^2}{2} u(t-1) - \frac{(t-\pi/2+\pi/2)^2}{2} u(t-\pi/2) - \cos(t-\pi/2+\pi/2)\, u(t-\pi/2)$$

$$f(t) = 2 - 2u(t-1) + \frac{1}{2}\left[(t-1)^2 + 2(t-1) + 1\right] u(t-1) - \frac{1}{2}\left[(t-\pi/2)^2 + 2(t-\pi/2) + \frac{\pi^2}{4}\right] u(t-\pi/2) - \sin\left(t - \frac{\pi}{2}\right) u(t-\pi/2)$$

$$f(t) \xrightarrow[\text{întârziere}]{\mathcal{L}} \frac{2}{p} - 2e^{-p}\cdot\frac{1}{p} + \frac{1}{2}e^{-p}\left(\frac{2}{p^3}+2\frac{1}{p^2}+\frac{1}{p}\right) -$$

$$-\frac{1}{2}e^{-\frac{\pi}{2}p}\left(\frac{2}{p^3}+\pi\cdot\frac{1}{p^2}+\frac{\pi^2}{4}\cdot\frac{1}{p}\right) - e^{-\frac{\pi}{2}p}\cdot\frac{1}{p^2+1}$$

ex 3 Să se determine funcțiile original ale căror transformate Laplace sunt:

a) $F(p) = \dfrac{p+1}{p^2+2p+\alpha}$ $\quad \alpha \in \mathbb{R}$

$$p^2+2p+\alpha = 0$$

$$\Delta = 4-4\alpha = 4(1-\alpha)$$

Cf pt. $\alpha<1 \Rightarrow \Delta>0 \Rightarrow p_{1/2} = \dfrac{-2\pm 2\sqrt{1-\alpha}}{2} = -1\pm\sqrt{1-\alpha}$

$$F(p) = \frac{p+1}{(p+1-\sqrt{1-\alpha})(p+1+\sqrt{1-\alpha})} = \frac{A}{p+1-\sqrt{1-\alpha}} + \frac{B}{p+1+\sqrt{1-\alpha}}$$

$$\Rightarrow p+1 = A(p+1+\sqrt{1-\alpha}) + B(p+1-\sqrt{1-\alpha})$$

$$\Rightarrow 1 = A+B \Rightarrow A = 1-B$$

$$1 = A + A\sqrt{1-\alpha} + B - B\sqrt{1-\alpha} \quad /\Rightarrow$$

$$\Rightarrow 1 = 1-B+\sqrt{1-\alpha}-B\sqrt{1-\alpha}+B-B\sqrt{1-\alpha} \Rightarrow 2B\sqrt{1-\alpha} = \sqrt{1-\alpha}$$

$$\Rightarrow 2B = 1 \Rightarrow B = \frac{1}{2} \qquad A = 1-\frac{1}{2} = \frac{1}{2}$$

$$F(p) = \frac{1}{2}\left(\frac{1}{p+1-\sqrt{1-\alpha}} + \frac{1}{p+1+\sqrt{1-\alpha}}\right) \xrightarrow{\mathcal{L}^{-1}}$$

$$\downarrow \mathcal{L}^{-1}$$

$$f(t) = \frac{1}{2}\left(e^{(-1+\sqrt{1-\alpha})t} + e^{(-1-\sqrt{1-\alpha})t}\right)$$

C II Pt $\alpha=1 \Rightarrow \Delta=0 \Rightarrow F(p)=\frac{p+1}{(p+1)^2}=\frac{1}{p+1} \xrightarrow{\mathcal{L}^{-1}} e^{-t}$

C III Pt $\alpha>1 \Rightarrow \Delta<0 \Rightarrow$

$$F(p)=\frac{p+1}{(p+1)^2+\alpha-1} \xrightarrow[\text{depl.}]{\mathcal{L}^{-1}} e^{-t}\cos\sqrt{\alpha-1}\,t$$

b) $F(p)=\frac{e^{-p}}{p^2-2p+5}+\frac{pe^{-2p}}{p^2+9}$

$$\frac{1}{p^2-2p+5}=\frac{1}{(p-1)^2+4} \xrightarrow[\text{depl.}]{\mathcal{L}^{-1}} f_1(t)=e^t\frac{1}{2}\sin 2t$$

$$F(p) \xrightarrow[\text{liu+int.}]{\mathcal{L}^{-1}} f(t)=\frac{1}{2}e^{t-1}\sin 2(t-1)+\cos 3(t-2)$$

ex4 Să se rezolve următoarea ecuație integrală

$$y(t)-\int_0^t y(\tau)(t-\tau)\,d\tau = 2-\frac{1}{2}t^2$$

Notăm $Y(p)=\mathcal{L}[y(t)](p)$

$$\int_0^t y(\tau)(t-\tau)\,d\tau = y(t)*t \xrightarrow[\text{conv.}]{\mathcal{L}} Y(p)\cdot\frac{1}{p^2}$$

$$2-\frac{1}{2}t^2 \xrightarrow[\text{liu.}]{\mathcal{L}} \frac{2}{p}-\frac{1}{2}\cdot\frac{2}{p^3}=\frac{2p^2-1}{p^3}$$

$$Y(t)-Y(p)\cdot\frac{1}{p^2}=\frac{2p^2-1}{p^3} \Rightarrow Y(t)\left(1-\frac{1}{p^2}\right)=\frac{2p^2-1}{p^3}$$

$$\Rightarrow Y(t)\cdot\frac{p^2-1}{p^2}=\frac{2p^2-1}{p^3} \Rightarrow Y(t)=\frac{2p^2-1}{p(p^2-1)}=\frac{A}{p}+\frac{B}{p+1}+\frac{C}{p-1}$$

$$2p^2-1 = A(p+1)(p-1)+Bp(p-1)+Cp(p+1)$$
$$= p^2(A+B+C)+p(-B+C)-A \Rightarrow \begin{cases} A+B+C=2 \\ C-B=0 \Rightarrow B=C \\ A=1 \end{cases}$$
$$\Rightarrow B=C=1$$

$$y(t) \xrightarrow{\mathcal{L}^{-1}} y(t) = A+Be^{-t}+Ce^{t} = 1+e^{-t}+e^{t}$$

ex5 Să se rezolve următoarea ecuație diferențială

a) $x''-2x'+5x = e^t\cos 2t \qquad x(0)=x'(0)=1$

Notăm $\mathcal{L}[x(t)](p) = X(p)$

$$x'(t) \xrightarrow[\text{orig.}]{\mathcal{L}\ \text{deriv.}} pX(p)-x(0) = pX(p)-1$$

$$x''(t) \xrightarrow[\text{orig}]{\mathcal{L}\ \text{deriv.}} p^2X(p)-px'(0)-x'(0) = p^2X(p)-p-1$$

$$e^t\cos 2t \xrightarrow[\text{deplas.}]{\mathcal{L}} \mathcal{L}[\cos 2t](p-1) = \frac{p-1}{(p-1)^2+4}$$

$$p^2X(p)-p-1-2pX(p)+2+5X(p) = \frac{p-1}{(p-1)^2+4}$$

$$X(p)(p^2-2p+5) = \frac{p-1}{(p-1)^2+4}+p-1$$

$$\Rightarrow X(p) = \frac{p-1}{(p^2-2p+5)^2}+\frac{p-1}{p^2-2p+5}$$

$$X(p) = \frac{p-1}{[(p-1)^2+4]^2}+\frac{p-1}{(p-1)^2+4} \xrightarrow{\mathcal{L}^{-1}} x(t) = f_1(t)+e^t\cos 2t$$

$$\frac{p-1}{[(p-1)^2+4]^2}=\frac{p-1}{(p-1)^2+4}\cdot\frac{1}{(p-1)^2+4}\xrightarrow{\mathcal{L}^{-1}}\text{conv.}\; e^t\cos 2t * \frac{1}{2}e^t\sin 2t$$

$$=\int_0^t e^\tau\cos 2\tau\cdot\frac{1}{2}\sin 2(t-\tau)\cdot e^{t-\tau}\,d\tau$$

$$=\frac{1}{2}e^t\int_0^t\cos 2\tau\cdot\sin 2(t-\tau)\,d\tau$$

$$=\frac{1}{2}e^t\int_0^t\cos 2\tau(\sin 2t\cos 2\tau-\sin 2\tau\cos 2t)\,d\tau$$

$$=\frac{1}{2}e^t\left[\sin 2t\int_0^t\underbrace{\cos^2 2\tau}_{\frac{1+\cos 4\tau}{2}}d\tau-\cos 2t\int_0^t\underbrace{\sin 2\tau\cos 2\tau}_{\frac{1}{2}\sin 4\tau}d\tau\right]$$

$$=\frac{1}{2}e^t\left[\frac{1}{2}\sin 2t\left(t+\frac{1}{4}\sin 4\tau\Big|_0^t\right)-\frac{1}{2}\cos 2t\left(-\frac{1}{4}\right)\cos 4\tau\Big|_0^t\right]$$

$$=\frac{1}{4}e^t\left[\sin 2t\left(t+\frac{1}{4}\sin 4t\right)+\frac{1}{4}\cos 2t(\cos 4t-1)\right]=f_1(t)$$

b) $y''+2y'+2y=\begin{cases}10\sin 2t & ;\ 0<t<\pi\\ 0 & ;\ t>\pi\end{cases}$ cu $\begin{matrix}y(0)=1\\ y'(0)=-5\end{matrix}$

Notăm $\mathcal{L}[y(t)](p)=Y(p)$

$$y'(t)\xrightarrow[\text{orig.}]{\mathcal{L}\ \text{deriv.}} pY(p)-y(0)=pY(p)-1$$

$$y''(t)\xrightarrow[\text{orig.}]{\mathcal{L}\ \text{deriv.}} p^2Y(p)-py(0)-y'(0)=p^2Y(p)-p+5$$

$$\begin{cases} 10\sin 2t & ;\ 0<t<\bar{\pi} \\ 0 & ;\ t>\bar{\pi} \end{cases} = 10\sin 2t\,(1-u(t-\bar{\pi}))$$

$u(t)$

$$= 10\sin 2t - 10\sin 2(t-\bar{\pi})\cdot u(t-\bar{\pi}) \longrightarrow$$

$$\xrightarrow[\text{întârz.}]{\mathcal{L}} 10\cdot\frac{2}{p^2+4} - 10e^{-p\bar{\pi}}\cdot\frac{2}{p^2+4} = \frac{20}{p^2+4}(1-e^{-p\bar{\pi}})$$

$$p^2Y(p) - p + 5 + 2pY(p) - 2 + 2Y(p) = \frac{20}{p^2+4}(1-e^{-p\bar{\pi}})$$

$$\Rightarrow Y(p)(p^2+2p+2) = \frac{20}{p^2+4}(1-e^{-p\bar{\pi}}) + p - 3$$

$$Y(p) = \frac{20}{(p^2+4)(p^2+2p+2)}(1-e^{-p\bar{\pi}}) + \frac{p-3}{p^2+2p+2}$$

$$Y(p) = \left(\frac{Ap+B}{p^2+4} + \frac{Cp+D}{p^2+2p+2}\right)(1-e^{-p\bar{\pi}}) + \frac{p-3}{(p+1)^2+1}$$

$$Y(p) = \left(A\cdot\frac{p}{p^2+4} + B\cdot\frac{1}{p^2+4} + C\cdot\frac{p+1-1}{(p+1)^2+1} + D\cdot\frac{1}{(p+1)^2+1}\right)$$

$$\cdot(1-e^{-p\bar{\pi}}) + \frac{p+1}{(p+1)^2+1} - \frac{4}{(p+1)^2+1}$$

$\mathcal{L}^{-1}$, deplas. întârz., liu.

$$y(t) = (A\cdot\cos 2t + B\cdot\frac{1}{2}\sin 2t + C\cdot e^{-t}\cos t - C\cdot e^{-t}\sin t +$$

$$+ D\cdot e^{-t}\sin t)\,u(t) - (A\cos 2(t-\bar{\pi}) + B\cdot\frac{1}{2}\sin 2(t-\bar{\pi}) +$$

$$+ Ce^{-(t-\bar{\pi})}\cos(t-\bar{\pi}) - Ce^{-(t-\bar{\pi})}\sin(t-\bar{\pi}) + De^{-(t-\bar{\pi})}\sin(t-\bar{\pi}))\cdot$$

$$\cdot u(t-\bar{\pi}) + (e^{-t}\cos t - 4e^{-t}\sin t)\,u(t)$$

ex6 Să se rezolve următoarea ecuație integro-diferențială

$$f''(t)+4\int_0^t e^{4(t-\theta)} f'(\theta)\,d\theta = \sin 5t \qquad f(0)=f'(0)=0$$

Notăm $\Phi(p)=\mathcal{L}[f(t)](p)$

$$f''(t) \xrightarrow[\text{orig.}]{\mathcal{L}} p^2\Phi(p)-pf(0_+)-f'(0)=p^2\Phi(p)$$

$$\int_0^t e^{4(t-\theta)} f'(\theta)\,d\theta = e^{4t} * f'(t) \xrightarrow[\text{conv.}]{\mathcal{L}} \frac{1}{p-4}\,p\Phi(p)$$

$$f'(t) \xrightarrow[\text{orig.}]{\mathcal{L}} p\Phi(p)-f(0)=p\Phi(p)$$

$$p^2\Phi(p)+4p\frac{1}{p-4}\Phi(p)=\frac{5}{p^2+25} \Rightarrow \Phi(p)\left(p^2+\frac{4p}{p-4}\right)=\frac{5}{p^2+25}$$

$$\Rightarrow \Phi(p)\cdot\frac{p^2(p-4)+4p}{p-4}=\frac{5}{p^2+25} \Rightarrow \Phi(p)=\frac{5(p-4)}{(p^2+25)p(p^2-4p+4)}$$

$$\Phi(p)=\frac{5(p-4)}{p(p^2+25)(p-2)^2}=\frac{A}{p}+\frac{Bp+C}{p^2+25}+\frac{D}{p-2}+\frac{E}{(p-2)^2}$$

$\mathcal{L}^{-1}$ | liv. dept.

$$f(t)=A+B\cos 5t+C\cdot\frac{1}{5}\sin 5t+D\cdot e^{2t}+E\cdot e^{2t}\cdot t$$

ex 7 Să se rezolve următoarea ecuație diferențială cu argumente întârziate

$$x''(t)+2x'(t-2)+x(t-4)=t \qquad x(0)=x'(0)=0$$

Notăm $X(p)=\mathcal{L}[x(t)](p)$

$$x(t-4) \xrightarrow[\text{întârz.}]{\mathcal{L}} e^{-4p}X(p)$$

$$x'(t-2) \xrightarrow[\text{întârz.}]{\mathcal{L}} e^{-2p}\mathcal{L}[x'(t)](p) \xrightarrow[\text{orig.}]{\text{deriv.}} e^{-2p}(pX(p)-x(0)) = e^{-2p}pX(p)$$

$$x''(t) \xrightarrow[\text{orig.}]{\mathcal{L}.\ \text{deriv.}} p^2X(p)-px(0)-x'(0)=p^2X(p)$$

$$p^2X(p)+2e^{-2p}pX(p)+e^{-4p}X(p)=\frac{1}{p^2}$$

$$\Rightarrow X(p)(p^2+2pe^{-2p}+e^{-4p})=\frac{1}{p^2} \Rightarrow X(p)(p+e^{-2p})^2=\frac{1}{p^2}$$

$$\Rightarrow X(p)=\frac{1}{p^2(p+e^{-2p})^2}=\frac{1}{p^4\left(1+\frac{e^{-2p}}{p}\right)^2}$$

$$\frac{1}{1+z}=\sum_{n\geq 0}(-1)^n z^n \xrightarrow{\text{deriv.}} -\frac{1}{(1+z)^2}=\sum_{n\geq 1}(-1)^n n z^{n-1}$$

$$\Rightarrow \frac{1}{(1+z)^2}=\sum_{n\geq 1}(-1)^{n+1} n z^{n-1}=\sum_{n\geq 0}(-1)^n (n+1) z^n$$

$$X(p)=\frac{1}{p^4}\cdot\sum_{n\geq 0}(-1)^n(n+1)\left(\frac{e^{-2p}}{p}\right)^n=\sum_{n\geq 0}(-1)^n(n+1)\frac{e^{-2np}}{p^{n+4}}$$

$$X(p) \xrightarrow[\text{+int.}]{\mathcal{L}^{-1}} x(t) = \sum_{n\geq 0} (-1)^n (n+1) \frac{(t-2n)^{n+3}}{\Gamma(n+4)}$$

$$x(t) = \sum_{n\geq 0} (-1)^n (n+1) \frac{(t-2n)^{n+3}}{(n+3)!}$$

ex8 Să se calculeze următoarea integrală cu ajutorul transformatei Laplace

$$y = \int_0^\infty \frac{x}{x^2+1} \sin 5x \, dx$$

$$y(t) = \int_0^\infty \frac{x \sin(tx)}{x^2+1} dx$$

$$\mathcal{L}[y(t)](p) = \int_0^\infty y(t) e^{-pt} dt = \int_0^\infty \left(\int_0^\infty \frac{x \sin(tx)}{x^2+1} dx \right) e^{-pt} dt =$$

$$= \int_0^\infty \frac{x}{x^2+1} \left(\underbrace{\int_0^\infty \sin(tx) e^{-pt} dt}_{\mathcal{L}[\sin tx](p) = \frac{x}{p^2+x^2}} \right) dx = \int_0^\infty \frac{x}{x^2+1} \cdot \frac{x}{x^2+p^2} dx$$

$$\frac{Ax+B}{x^2+1} + \frac{Cx+D}{x^2+p^2} = \frac{x}{x^2+1} \cdot \frac{x}{x^2+p^2} \Rightarrow$$

$$\Rightarrow (Ax+B)(x^2+p^2) + (Cx+D)(x^2+1) = x^2$$

$$x^3(A+C) + x^2(B+D) + x(p^2A+C) + Bp^2 + D = x^2$$

$$\begin{cases} A=-C \\ B+D=1 \\ p^2A=-C \\ Bp^2+D=0 \Rightarrow D=-Bp^2 \end{cases} \Rightarrow A=C=0$$

$$B-Bp^2=1 \Rightarrow B=\frac{1}{1-p^2}=\frac{-1}{p^2-1}$$

$$D=1+\frac{1}{p^2-1}=\frac{p^2}{p^2-1}$$

$$\mathcal{L}[y(t)](p)=\int_0^\infty \frac{x}{x^2+1}\cdot\frac{x}{x^2+p^2}\,dx=\int_0^\infty \frac{-1}{(p^2-1)(x^2+1)}+\frac{p^2}{(p^2-1)(x^2+p^2)}\,dx$$

$$=-\frac{1}{p^2-1}\operatorname{arctg}x\Big|_0^\infty+\frac{p^2}{p^2-1}\cdot\frac{1}{p}\operatorname{arctg}\frac{x}{p}\Big|_0^\infty$$

$$=-\frac{1}{p^2-1}\cdot\frac{\pi}{2}+\frac{p}{p^2-1}\cdot\frac{\pi}{2}=\frac{\pi}{2}\cdot\frac{1}{p^2-1}(p-1)=\frac{\pi}{2}\cdot\frac{1}{p+1}$$

$$y^2(t)=\mathcal{L}^{-1}\left[\frac{\pi}{2}\cdot\frac{1}{p+1}\right](t)=\frac{\pi}{2}\cdot e^{-t}$$

$$y=y(5)=\frac{\pi}{2}e^{-5}$$